THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x + \frac{1}{x} = 4$, then $x^3 + \frac{1}{x^3} =$ 1

(a) 51 **(b)** 52 **(c)** 54 **(d)** 53

2. The value of 99^3 is: 1

(a) 970298 **(b)** 970300 **(c)** 970299 **(d)** 970301

3. If $a + b + c = 0$, then $a^3 + b^3 + c^3$ for $a = 4$, $b = -1$, $c = -3$ equals: 1

(a) 37 **(b)** 36 **(c)** 38 **(d)** 35

4. If $a + b + c = 12$ and $a^2 + b^2 + c^2 = 50$, then $ab + bc + ca =$ 1

(a) 49 **(b)** 47 **(c)** 46 **(d)** 48

5. If $x^2 + \frac{1}{x^2} = 23$, then $\left(x + \frac{1}{x}\right)^2 =$ 1

(a) 24 **(b)** 25 **(c)** 26 **(d)** 27

6. $27x^3 - 8$ factorises as: 1

(a) $(3x - 2)(9x^2 + 6x + 4)$ **(b)** $(3x - 2)(9x^2 - 6x + 4)$ **(c)** $(3x - 2)^3$ **(d)** $(3x + 2)(9x^2 - 6x + 4)$

7. If $x + y = 6$ and $xy = 8$, then $x^3 + y^3 =$ 1

(a) 74 **(b)** 72 **(c)** 71 **(d)** 73

8. If $x - y = 1$ and $xy = 12$, then $x^3 - y^3 =$ 1

(a) 38 **(b)** 36 **(c)** 37 **(d)** 39

9. If $x + \frac{1}{x} = 5$, then $x^3 + \frac{1}{x^3} =$ 1

(a) 112 **(b)** 109 **(c)** 110 **(d)** 111

10. The value of 102^3 is: 1

(a) 1061209 **(b)** 1061207 **(c)** 1061210 **(d)** 1061208

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of 97^3 using an identity. 2

12. Factorise $27x^3 + 8$. 2

13. If $x + y = 6$ and $xy = 5$, find $x^3 + y^3$. 2

14. Find the value of 105^3 using an identity. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $x + \frac{1}{x} = 9$, find $x^3 + \frac{1}{x^3}$. 3

16. If $a + b + c = 0$, show that $a^3 + b^3 + c^3 = 3abc$ for $a = 3$, $b = 5$, $c = -8$. 3

17. Factorise $x^3 - 216$. 3

OR

If $x + \frac{1}{x} = 10$, find $x^3 + \frac{1}{x^3}$. 3

18. If $a + b + c = 0$, show that $a^3 + b^3 + c^3 = 3abc$ for $a = -2$, $b = 9$, $c = -7$. 3

OR

Factorise $125x^3 - 8$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. If $x - y = 5$ and $xy = 14$, find $x^2 + y^2$, $x^3 - y^3$ and $(x + y)^2$. 5

OR

If $a + b + c = 12$ and $ab + bc + ca = 47$, find $a^2 + b^2 + c^2$ and, if $abc = 60$, evaluate $a^3 + b^3 + c^3$. 5

20. If $x - y = 4$ and $xy = 45$, find $x^2 + y^2$, $x^3 - y^3$ and $(x + y)^2$. 5

OR

If $a + b + c = 10$ and $ab + bc + ca = 31$, find $a^2 + b^2 + c^2$ and, if $abc = 30$, evaluate $a^3 + b^3 + c^3$. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND